# LAB 11 AND 12 RESULTS

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# OVERVIEW

**Objective**

- Measure trace resistance for multiple different widths
- Utilize the 2-wire vs 4-wire measurement methods
- Observe the limits of the traces by sending excessive current

**Measurement Differences**

- Compare the differences between the 2-wire and 4-wire measurements
- Compare to calculated estimated values for each trace

# TRACE RESISTANCE FINDINGS

6 mil, 10 mil, 20 mil, and 100 mil widths were tested. A power supply and digital multimeter were used. The 2-wire method utilized only the digital multimeter and the 4-wire method utilized both for most accurate measurements. It is clear that the 4-wire method most closely aligns with the estimated trace resistance values. A constant current of 1 A is used.

| Trace Width | Estimated Trace Resistance |
|---|---|
| 6 mil | 0.083 Ω |
| 10 mil | 0.05 Ω |
| 20 mil | 0.025 Ω |
| 100 mil | 0.005 Ω |

| Trace Width | 2-Wire | 4-Wire |
|---|---|---|
| 6 mil | 0.22 Ω | 0.075 Ω |
| 10 mil | 0.17 Ω | 0.040 Ω |
| 20 mil | 0.15 Ω | 0.019 Ω |
| 100 mil | 0.13 Ω | 0.004 Ω |

*It is evident that the 4-wire method performs most accurately when compared to the estimated values. The 2-wire method is much larger than what is predicted.*

# TRACE LIMIT FINDINGS

The 6 mil and 20 mil traces were tested to see how much current they could handle.

| Trace Width | Gets Warm (A) | Gets Hot (A) | Gets Very Hot (A) | Burns (A) |
|---|---|---|---|---|
| 6 mil | 1.25 A | 2.75 A | 3.50 A | 4.56 A |
| 20 mil | 3.50 A | 4.75 A | 6.00 A | 10 A |

*It is evident that the 20-mil trace performs far better than the 6 mil. It took the maximum current output of the power supply to burn the trace. The 6-mil trace broke down at 4.56 A.*

- *The 6-mil traced conducted ~0.8 V as it burned (Significantly higher than the 75.27 mV it conducted at 1 A)*
- *The 20-mil trace conducted ~381.7 mV as it burned (Much higher than the 3.92 mV it conducted at 1 A)*

# FULL MEASUREMENTS

1 oz : thickness = $34 \mu m \approx 1.4$ mils

1 oz copper: $R_{\square} = \frac{1.7 \times 10^{-8}\, \Omega \cdot m}{34 \times 10^{-6}\, m} = 0.5\, m\Omega/\square$

$$R = \rho \frac{len}{Area} = \frac{\rho}{t} \times \frac{len}{W} = \frac{\rho}{t} \times n$$

$$= R_{\square} \times n = 0.5\, m\Omega \times n$$

$$\frac{1''}{0.006''} = 166.67$$

$$\frac{1''}{0.010''} = 100$$

$$\frac{1''}{0.020''} = 50$$

$$\frac{1''}{0.100''} = 10$$

| Line Width | Estimate | 2-Wire | 4-Wire with 1 A |
|---|---|---|---|
| 6 mil | 0.083 Ω | 0.22 Ω | 0.075 Ω |
| 10 mil | 0.05 Ω | 0.17 Ω | 0.040 Ω |
| 20 mil | 0.025 Ω | 0.15 Ω | 0.019 Ω |
| 100 mil | 0.005 Ω | 0.13 Ω | 0.004 Ω |

$$R_{6\,mil} = (0.5\, m\Omega)(166.67) = 0.083\, \Omega$$

$$R_{10\,mil} = (0.5\, m\Omega)(100) = 0.05\, \Omega$$

$$R_{20\,mil} = (0.5\, m\Omega)(50) = 0.025\, \Omega$$

$$R_{100\,mil} = (0.5\, m\Omega)(10) = 0.005\, \Omega$$

4 - Wire:

$$R_{Trace} = \frac{V_{meas.}}{I_{through}}$$

$$R_{Trace_{6\,mil}} = \frac{75.27\, mV}{1\, A} = 0.075\, \Omega$$

$$R_{Trace_{10\,mil}} = \frac{39.71\, mV}{1\, A} = 0.040\, \Omega$$

$$R_{Trace_{20\,mil}} = \frac{18.74\, mV}{1\, A} = 0.019\, \Omega$$

$$R_{Trace_{100\,mil}} = \frac{3.92\, mV}{1\, A} = 0.004\, \Omega$$